Scaling Genetic Testing Policies with Reinforcement Learning

When exact dynamic programming breaks, RL keeps going

PPO **Bayesian belief MDP** **N = 4–15 members** **1–3 genes**

Rajhansini · June 2026

The Bottleneck

- Sequential genetic testing: choose *who to test next* or *stop*
- Belief state tracks carrier probabilities via Mendelian inheritance
- **Exact backward induction** gives optimal policies — but state space grows as 3^N

256

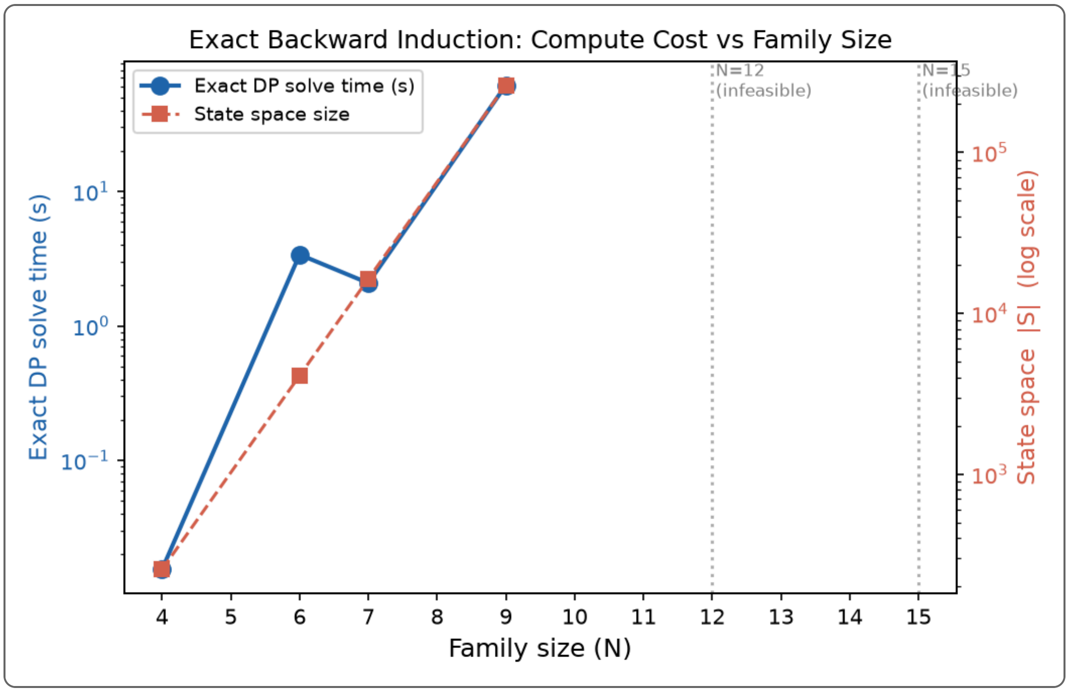
states at N=4

262K

states at N=9

1B+

states at N=15



Exact solve: 0.02 s (N=4) → 62 s (N=9) → infeasible (N≥12)

Our Approach

Pedigree + MDP
Mendelian genetics
test / stop actions

→

Gym Environment
Belief-state obs
validated vs exact

→

PPO Agent
MLP actor-critic
~300K timesteps

→

Evaluate
vs exact V*
vs myopic / random

Environment details

- ✓ Observation: carrier beliefs per individual × gene
- ✓ Reward: outcome utility − testing costs (fixed + variable)
- ✓ Allele frequency 0.1 (BRCA-style); sensitivity 0.05–0.30
- ✓ Bayesian updates validated against hand-computed posteriors

Why RL?

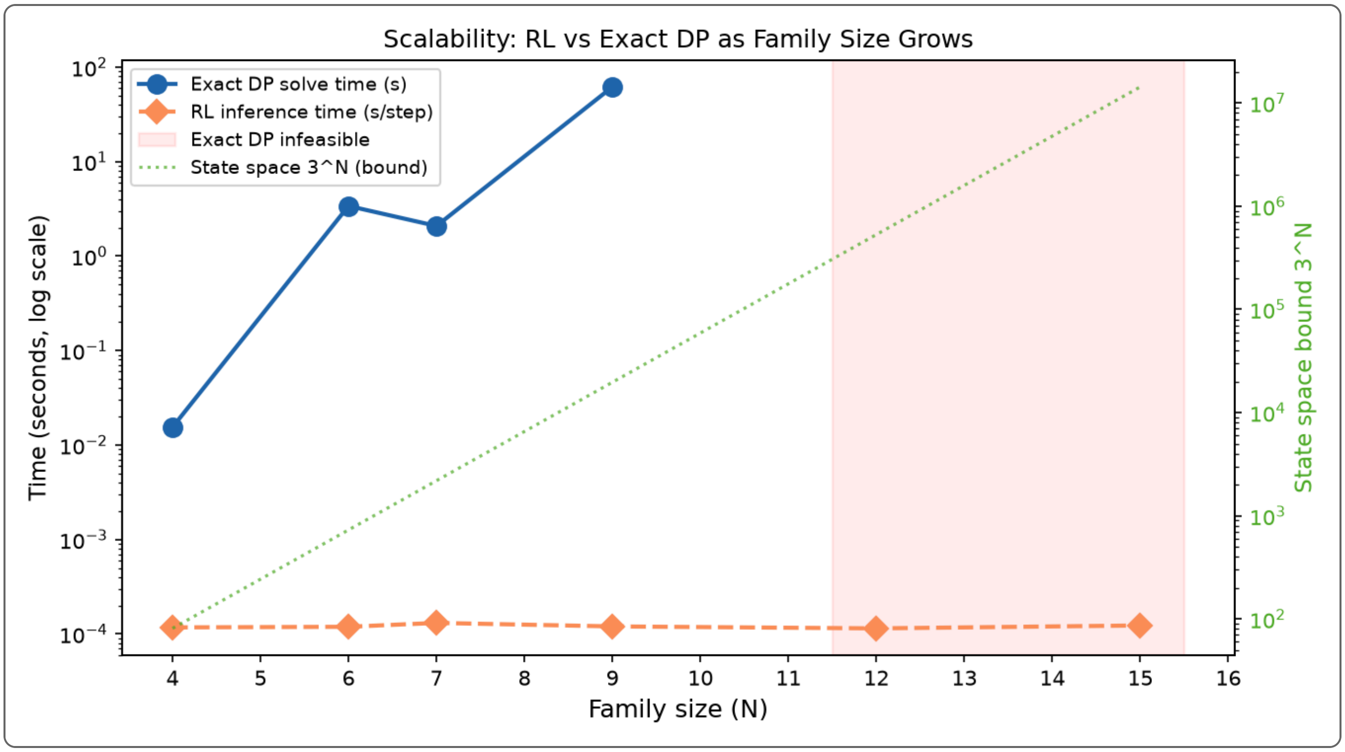
- Inference is **~0.12 ms/step** — flat across N

- Training cost is one-time; deploy policy on any pedigree size
- No enumeration of 3^N genotype configurations

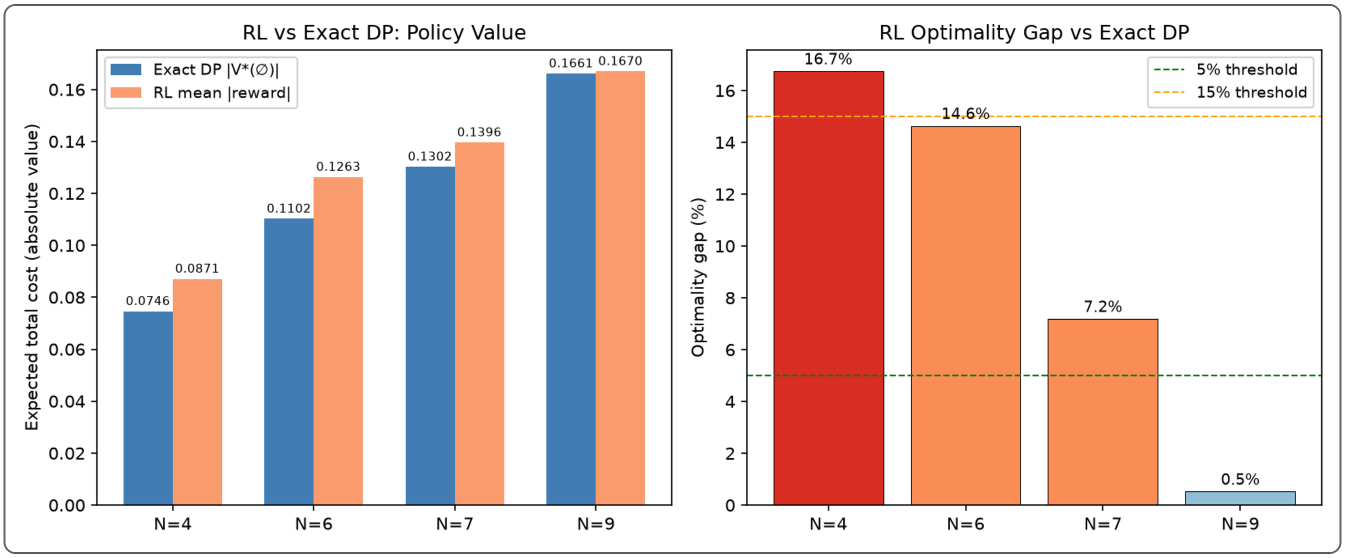
Week 3 — RL vs Exact DP

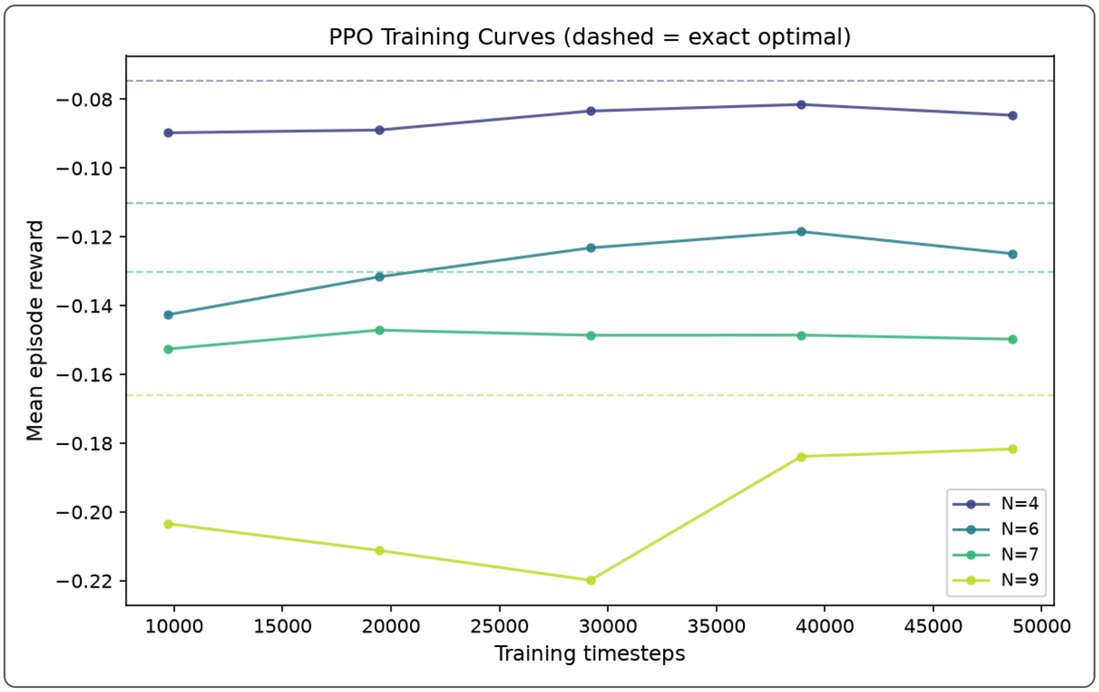
N	Exact V*	RL mean	Gap	Exact time
4	−0.075	−0.087	16.7%	0.02 s
6	−0.110	−0.126	14.6%	3.4 s
7	−0.130	−0.140	7.2%	2.1 s
9	**−0.166**	**−0.167**	**0.5%**	**62 s**
12	—	—	—	infeasible
15	—	—	—	infeasible

RL inference: ~0.12 ms/step at all N (500× faster than exact solve at N=9)


Scalability: RL vs Exact DP as Family Size Grows
Exact DP solve time (s)
RL inference time (s/step)
Exact DP infeasible
State space 3^N (bound)
Time (seconds, log scale)
State space bound 3^N
Family size (N)

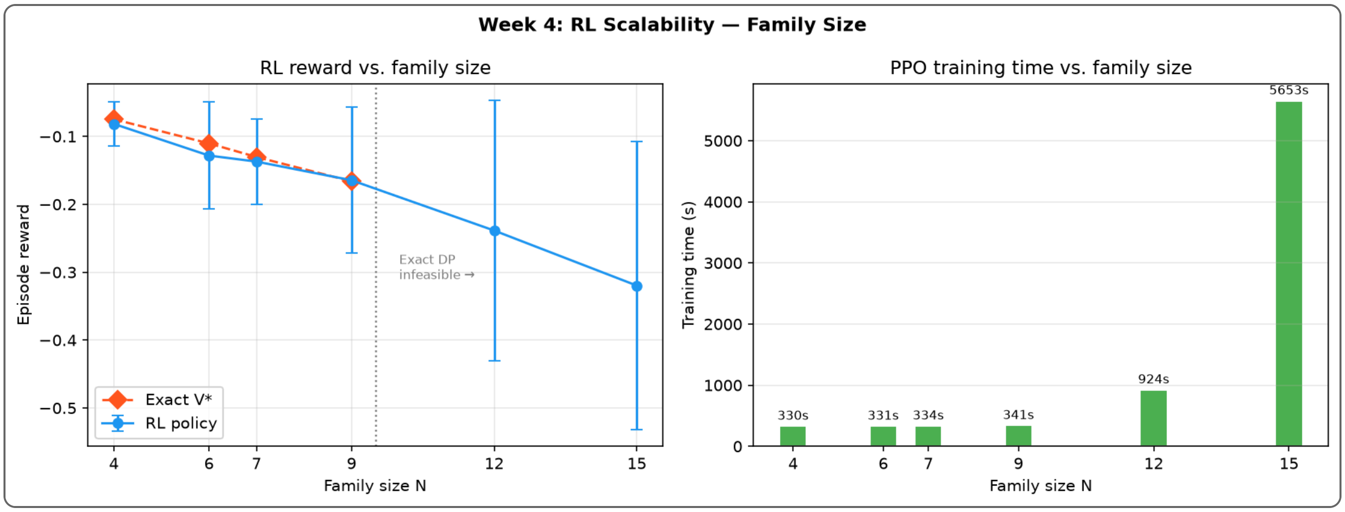
RL Converges Toward Optimum





Gap drops from ~17% (N=4) to **<1%** (N=9) as state complexity increases — PPO benefits from richer decision structure

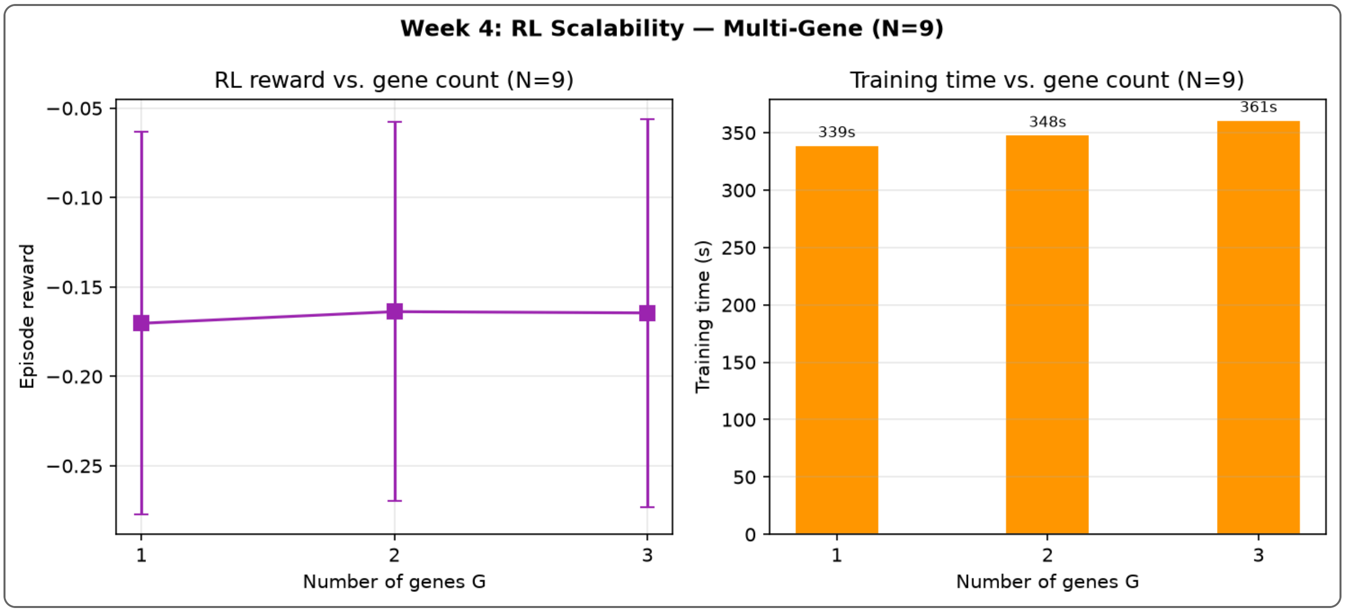
Week 4 — Scaling Past Exact DP



N	State bound	RL reward	Gap vs exact
4	256	−0.081	9.2%
6	4,096	−0.128	16.3%
7	16,384	−0.137	5.2%
9	262,144	−0.165	1.0%
12	16.8M	−0.239	—
15	1.07B	−0.320	—

RL trains successfully where exact DP cannot run.
Training time: ~5.5 min (N=9) → ~94 min (N=15).

Multi-Gene Families (N = 9)



- Extended obs space: beliefs per individual × gene
- 1 gene (G=1): reward −0.170, gap 2.5% vs exact
- 2 genes (G=2): reward −0.164 — no degradation
- 3 genes (G=3): reward −0.164 — stable

~6 min

train time per config

0.15 ms

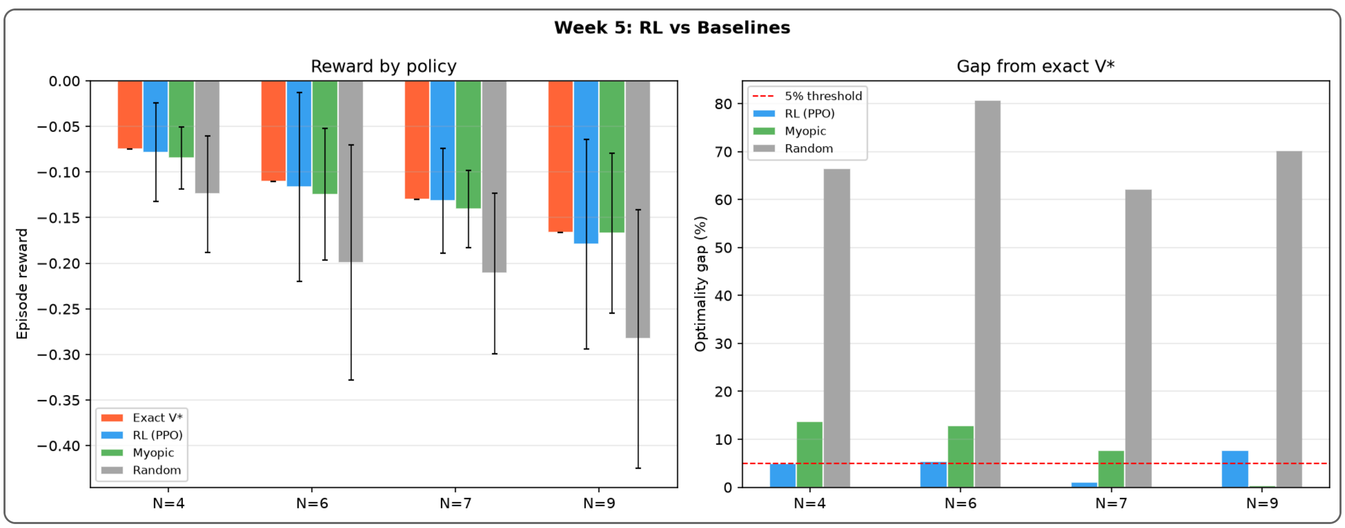
inference / step

RL handles multi-gene pedigrees without exact ground truth — path to 5–20 gene families.

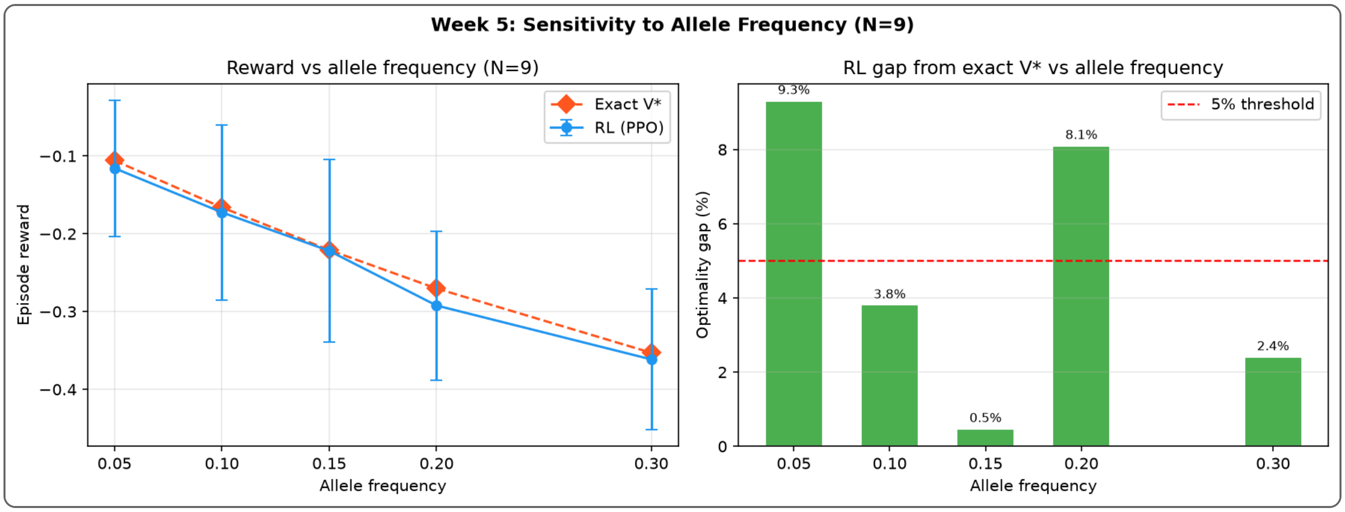
Week 5 — RL vs Baselines

N	Exact	Random	Myopic	RL
4	−0.075	−0.124	−0.085	**−0.078**
6	−0.110	−0.199	−0.124	**−0.116**
7	−0.130	−0.211	−0.140	**−0.132**
9	−0.166	−0.283	−0.167	−0.179

RL beats **random** by 60–80% gap reduction.
RL beats **myopic** at N=4,6,7; competitive at N=9.
Myopic over-tests at N=7 (6.1 tests vs RL 5.2).



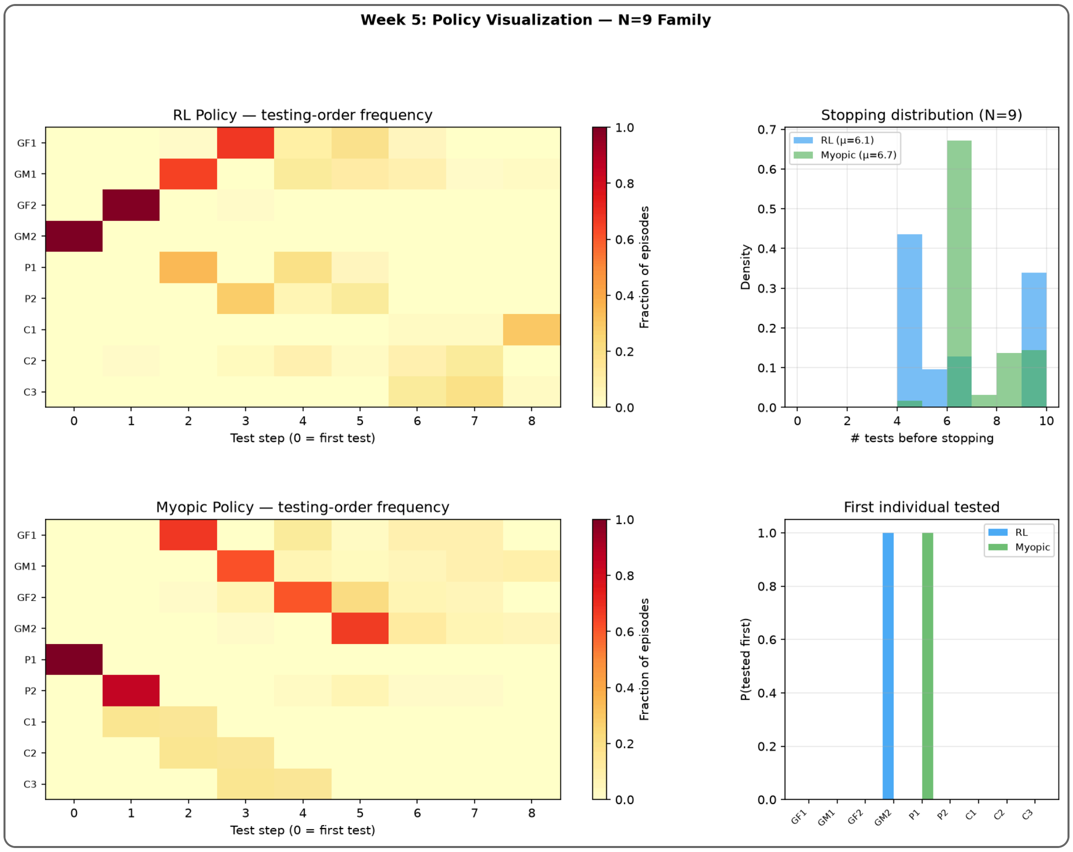
Sensitivity to Allele Frequency



Freq	Exact V*	RL	Gap
0.05	−0.106	−0.116	9.3%
0.10	−0.166	−0.172	3.8%
0.15	−0.221	−0.222	**0.5%**
0.20	−0.270	−0.292	8.1%
0.30	−0.353	−0.361	2.4%

RL tracks exact optimum across BRCA-relevant allele frequencies without re-architecting the policy.

What Does the Learned Policy Do?



- **Testing order:** RL prioritizes high-information individuals (parents before children)
- **Stopping:** RL averages **6.05 ± 2.21** tests vs myopic **6.70 ± 1.20**
- **Efficiency:** fewer wasted tests while maintaining reward quality
- Policy structure resembles exact optimal ordering on N=9 three-generation pedigree

2,000 evaluation episodes · trained policy: `rl_policy_N9.pt`

Key Takeaways

1. Near-optimal where we can verify

RL matches exact DP within <1% at N=9; gaps 0.5–9% across allele frequencies

2. Scales where exact methods fail

Trained policies for N=12 (16M states) and N=15 (1B states); multi-gene families at N=9

3. Beats simple heuristics

Consistently outperforms random; beats myopic on small/medium families; more efficient stopping

Exact DP gives the certificate · RL gives the scalable deployable policy

Future Work

Near term

- GNN state encoder for pedigree structure (swappable module)
- Scale to 5–20 genes on larger pedigrees
- Compare RL to ABCD-16 ADP bounds from exact replication package
- Transfer learning: train on N=15, fine-tune on new topologies

Longer term

- Real clinical pedigrees (BRCA1/2 cascade screening)
- Constrained policies (cost budgets, test limits)
- Uncertainty quantification on policy value
- Integration with existing Gurobi ADP pipeline

All code, results, and 10 figures: `replication_16features/artifacts/`

Thank You

Questions?

Navigate: ← → arrows · Fullscreen: F · Overview: Esc · Speaker notes: S

Speaker notes